

## Unit 3, Lesson 7: Dividing Numbers Expressed in Scientific Notation

### Benchmarks

**MA.8.NSO.1.5** Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

**MA.8.NSO.1.6** Solve real-world problems involving operations with numbers expressed in scientific notation.

### Learning Targets

- ☐ I can divide numbers expressed in scientific notation.
- ☐ I can solve real-world problems involving operations with numbers expressed in scientific notation.

3.7.1 Warm-Up: Toilet Paper Math

The table shows information from four different toilet paper roll packages.

|                                                                                    |                                                                                     |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 18 Mega Rolls = 72 Regular Rolls                                                   | 24 Family Mega Rolls = 123 Regular Rolls                                            |
|   |   |
| 30 Mega Rolls = 120 Regular Rolls                                                  | 18 Family Mega Rolls = 90 Regular Rolls                                             |
|  |  |

1. How many regular rolls are equal to 1 mega roll?
2. Explain why determining the number of Regular Rolls packs that are equal to 1 Family Mega Rolls pack would require you to estimate.
3. Write a question about the toilet paper rolls that could be asked to a 6<sup>th</sup> grader.

### 3.7.2 Exploration: Making a Conjecture for How to Divide Numbers Expressed in Scientific Notation

Work with your partner to complete the problem in each table. First, determine the answer to the division problem in the first row, using whole numbers. Then, convert the whole numbers to scientific notation in the second row.

| 1.                  | Problem       | Answer            |
|---------------------|---------------|-------------------|
| Whole Numbers       | $600 \div 25$ |                   |
| Scientific Notation |               | $2.4 \times 10^1$ |

| 2.                  | Problem        | Answer            |
|---------------------|----------------|-------------------|
| Whole Numbers       | $4500 \div 30$ |                   |
| Scientific Notation |                | $1.5 \times 10^2$ |

| 3.                  | Problem                 | Answer            |
|---------------------|-------------------------|-------------------|
| Whole Numbers       | $7,000,000 \div 28,000$ |                   |
| Scientific Notation |                         | $2.5 \times 10^2$ |

| 4.                  | Problem                  | Answer               |
|---------------------|--------------------------|----------------------|
| Whole Numbers       | $70,000 \div 28,000,000$ |                      |
| Scientific Notation |                          | $2.5 \times 10^{-3}$ |

| 5.                  | Problem               | Answer             |
|---------------------|-----------------------|--------------------|
| Whole Numbers       | $90,000 \div 180,000$ |                    |
| Scientific Notation |                       | $5 \times 10^{-1}$ |

6.    What do you notice about the problems, using scientific notation and their corresponding answer?
7.    Make a conjecture about how to divide numbers written in scientific notation.
8.    Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

| My Summary of Their Conjecture | Initials |
|--------------------------------|----------|
|                                |          |
|                                |          |

9.    Revisit your conjecture with your partner and resolve any differences if necessary.

**3.7.3 Guided Instruction: Dividing Numbers Written in Scientific Notation**

1. Gizmo Gadgets produced 50,000 widgets last month. They shipped them out to 2,500 different stores.

a. How many widgets did each store receive? Write your answer in scientific notation.

- b. Determine if the following equation is true. Support your answer using mathematics.

$$2 \times 10^1 = \frac{(5 \times 10^4)}{(2.5 \times 10^3)}$$

- c. Explore what happens when the dividend is less than the divisor by completing the problem.

$$\frac{(7 \times 10^4)}{(2.8 \times 10^6)} = \text{_____} \times 10^{4-6} = 2.5 \times 10$$

- d. Discuss with your partner why the answer to Part C is not a negative number and has a negative exponent.

- e. Solve by filling in the blanks.

$$\frac{(-7 \times 10^4)}{(2.8 \times 10^6)} = \text{_____} \times 10^{4-6} = -2.5 \times 10$$

- f. Discuss with your partner why the answer to Part E is a negative number.

2. Mara and Colin were given the following problem. Their work is shown.

The distance from the sun to the Earth is about 150,000,000,000 meters. The speed of light is  $(3 \times 10^9)$  meters per second. How many seconds does it take for light from the sun to reach the Earth?

| Mara's Work                                                                                                                                   | Colin's Work                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= \frac{(1.5)}{(3)} \times \frac{(10^{11})}{(10^9)}$ $= (.5 \times 10^2)$ $= (5 \times 10^2)$ | $\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= 5 \times 10^1$ |

- a. Place your initials by the student's name who has the correct answer in the table.
- b. Circle and correct the error made by the student with the incorrect answer.
- c. Write a note to the person who is incorrect, explaining their mistake.

**Means**

**I**

**Start**

**To**

**Acquire**

**Knowledge**

**Experience**

**Skills**

### 3.7.4 Guided Practice: Finding Quotients

Find the quotient for each.

1.  $\frac{(5.5 \times 10^3)}{(2.75 \times 10^4)}$

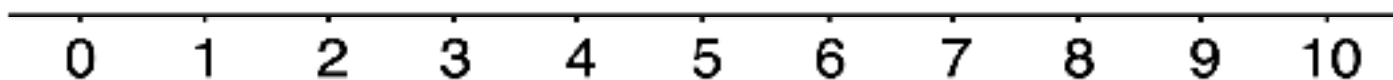
2.  $\frac{(7.2 \times 10^{24})}{(8 \times 10^{20})}$

3.  $\frac{(2.16 \times 10^{-3})}{(2 \times 10^{-6})}$

4.  $\frac{(9.87 \times 10^{-4})}{(4.7 \times 10^{-8})}$

### 3.7.5 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of dividing numbers expressed in scientific notation.



2. Complete the statement.

I chose this emoji because \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_